

Experiment 2

CHARACTERISTICS OF GUNN DIODE

Experiment 2

Objective:

- Study the effect of bias variation on diode current and frequency.
- The effect of mechanical tuning on oscillator output power.
- Measurement of pushing figure.

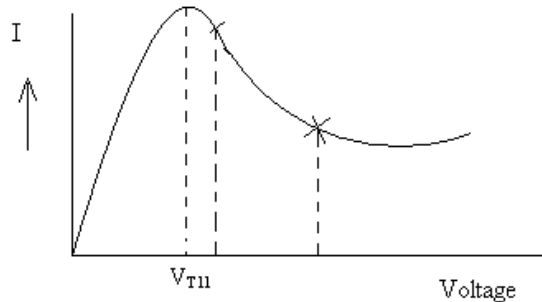


Fig 1. Voltage-current characteristic of a Gunn diode.

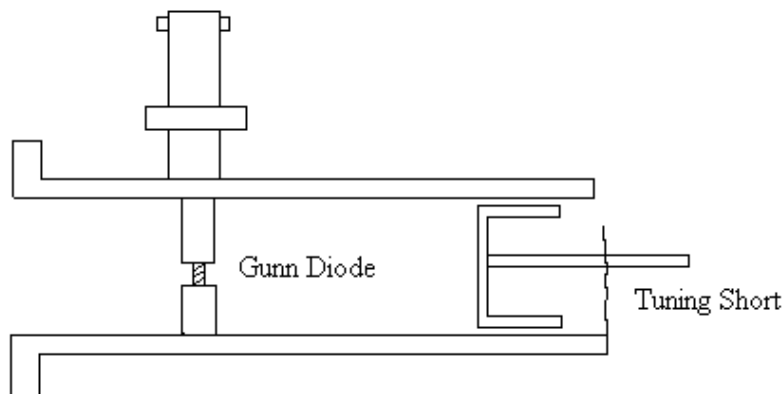


Fig 2. A tunable Gunn oscillator.

Theory:

Gunn diode:

Gunn diode is an example of a transferred electron device (TED). In the Gunn diode, two regions are heavily *n*-doped on each terminal. A thin layer of lightly *n*-doped material in the middle is called the active region. It therefore does not conduct in only one direction and cannot rectify alternating current like a conventional diode with a *pn* junction. Its largest use is in electronic oscillators to generate microwave signal. The working principle is based on the "Gunn effect" discovered in 1962 by physicist J. B. Gunn. Some group III-V semiconductor materials like GaAs behaves as a normal semiconductor material at low electric field but shows negative resistance at high electric field (hot electron). Negative resistance is a special property of the Gunn diode in which

an increase in voltage across the device's terminals results in a decrease in current. Unlike a positive resistance that consumes power from current passing through it, a negative resistance produces power, amplifying it.

GaAs has two conduction bands, known as lower and upper valleys. Electrons in the lower valley have high mobility, small effective mass, and low density of state. When a voltage is applied to the device, the electrical gradient is largest across the thin middle layer. If the voltage is increased, the current through the layer first increases, but eventually, at higher field, more electrons are transferred to the upper valley increasing its resistivity and causing the current to fall. The conductive property of the middle layer is altered. Thus, a Gunn diode has a region of negative differential mobility which results in different slope in its current-voltage characteristic curve as shown in Fig. 1. This negative resistance property allows it to amplify, functioning as a radio frequency amplifier, or to become unstable and oscillate when it is biased with a DC voltage.

The negative differential mobility gives rise to a space charge instability and is responsible for the formation of high field domains in Gunn devices when a suitable DC voltage is applied across the diode. High field domains originate near the cathode and drift across the sample towards the anode. The transit time of these 'domains' determines the periodicity of the external current waveform when the circuit is resistive. When the diode is placed in a resonant circuit, various modes of operation are possible depending upon the relationship between transit time frequency and the frequency of the resonant circuit.

Gunn diode oscillator used in the laboratory employs a device mounted in the waveguide. The schematic of an X-band waveguide oscillator is shown in Fig. 2. In this oscillator, the cavity consists of a waveguide section with a movable short circuit at one end and the diode at the other end. The movable short can be continuously moved by turning the tuning knob and thus changing the oscillation frequency.

Procedure:

1. Set up the experiment as shown in Fig. 3 with the attenuator at maximum attenuation.
2. Connect the power supply to Gunn oscillator.
3. Connect the bolometer and the laptop.
4. Turn on the power supply with its main switch (**Caution:** Never exceed 10 V DC. It is recommended to set the DC bias voltage to zero before switching on the DC source).

Voltage versus current characteristic:

- i) Measure the diode current as a function of bias voltage (4 – 9.0 V). Do not exceed 9.5 volts. Please note that the actual DC voltage drop across the Gunn diode is not accessible. Therefore, record and plot the voltages displayed on the supply meter.
- ii) Plot bias voltage V and diode current I .
- iii) Determine the threshold voltage V_{th} and the corresponding maximum current.

Frequency tuning:

1. Bias the Gunn oscillator at 9 volts.
2. Tune the Gunn oscillator frequency to approximately 9 GHz by turning the tuning knob. Measure the frequency and power output.
3. Tune the Gunn oscillator at various frequencies and measure the corresponding output power (8-10 readings over a wide frequency band).
4. Note down the screw reading for each frequency point.
5. Plot power output versus frequency.

Pushing Figure:

Pushing figure is a measure of frequency sensitivity against variation in dc voltage for an electronically tuned oscillator. It is defined as the incremental change in frequency with the increasing voltage. Its unit is MHz/Volt. Pushing figure may be either positive or negative.

Tune the Gunn oscillator frequency to approximately 10 GHz. Note down the frequency. Adjust the bias at 8.0 V. Now vary the bias by about 0.5 V and measure the frequency again. Determine the pushing figure.

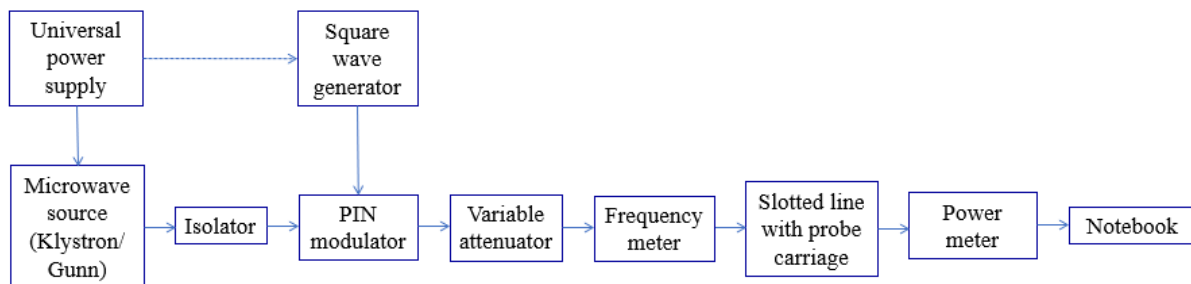


Fig. 3 Measurement set-up (the PIN modulator remains off).